AYON ROY

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SUMMARY AND RESEARCH INTERESTS

Computer Science researcher and masters student in Cyber-Physical Systems and Services at Université Jean Monnet, France. I completed my BSc in Computer Science and Engineering at BRAC University, Bangladesh, with a CGPA of 3.90/4.00 (Highest Distinction). My goal is to build relaiable systems that can make decisions without depending entirely on LLMs. My experience includes AI and Guardrail Agent research, telecommunications, NLP data prweprocessing and Bangla machine translation. I am seeking for PhD opportunities in NLP, Agentic Artificial Intelligence, Robotics Simulation, with applications to real-world systems.

EDUCATION

09/2026–Present
Master of Science (M1) in Cyber-Physical Systems and Services (CPS2)
Université Jean Monnet, Saint-Étienne, France

10/2020–02/2025
Bachelor of Science in Computer Science and Engineering
BRAC University, Dhaka, Bangladesh
136 credits (four years) **CGPA: 3.90/4.00 (Highest Distinction)**

07/2017–07/2019
Higher Secondary School Certificate
Cumilla Victoria Govt College, Cumilla, Bangladesh
GPA: 5.0/5.0

LANGUAGE PROFICIENCY

IELTS
Band 7.5
Listening: 8.0 • Reading: 7.0 • Writing: 7.0 • Speaking: 7.0
Valid until: 27 September 2027

THESIS

Advancing Bangla Machine Translation Through Informal Datasets
Available at: doi.org/10.48550/arXiv.2512.13487

Created an informal Bangla–English sentence dataset and trained hybrid and deep-learning models to improve translation quality for low-resource and informal language. The dataset was manually created and translated from social-media conversations and made publicly available for further research.

RESEARCH AND PUBLICATIONS

Adaptive Alarm Threshold Prediction in 4G Mobile Networks: A Percentile-Guided Deep Learning Framework with Interpretable Outputs
Available at: doi.org/10.48550/arXiv.2605.00838 *(Under peer review at the Journal of Computer Networks)*

Presents a deep-learning approach to predicting suitable alarm thresholds for 4G mobile networks based on network behaviour. The model makes alarm settings more adaptive and accurate while using fewer parameters than existing models.

Beyond Pandas: A Dask-Based Parallel Pipeline for Large-Scale Synthetic Data Generation
Accepted at the ICCA conference; forthcoming in the ACM Digital Library.

Presents a Dask-based parallel approach for generating large volumes of synthetic financial data more efficiently than Pandas, reducing execution time and memory usage while maintaining statistical quality.

Layerwise Sparsity Allocation in Convolutional Neural Networks via a Linear-Chain CRF Energy Model
Accepted at the ICCA conference; forthcoming in the ACM Digital Library.

Proposes a CNN-pruning method that determines how much to remove from each layer while accounting for dependencies between neighbouring layers. The Chain-CRF method improves performance at high compression levels while preserving model accuracy.

Probe-Reliability and Architectural Compensation in CNN Channel Pruning
Accepted at the MIET conference; forthcoming in a Scopus-indexed Springer Nature book.

Investigated reliable methods for identifying removable CNN channels while preserving model performance, demonstrating that channel-importance measures are architecture-dependent and proposing Group-Taylor scoring to improve pruning decisions in depthwise and SE-gated networks.

RELEVANT PROJECTS

Academic

- **Online Payments Fraud Detection:** Classified fraudulent online transactions using K-nearest neighbours, logistic regression, and naive Bayes.
- **Detecting Sarcasm in English:** Classified slang and sarcasm in English sentences using GRU and LSTM models.
- **Blind Stick with Morse-Code Response and GSM:** Built an assistive smart-stick prototype using Arduino and multiple sensors.

Industrial

- **2G Dimensioning Busy-Hour Traffic Prediction:** Developed a SARIMA-based system with a front end that uses cleaned historical data and filters to forecast traffic, improving service quality and reducing call drops by approximately 75%.
- **Fluctuating Cell Detection and Responsible-Team Notification:** Built a Python-based automated system that detects fluctuating network cells every six hours, flags issues based on defined thresholds, and emails the responsible teams, enabling faster responses and reducing operational costs.

PROFESSIONAL EXPERIENCE

06/2025–06/2026
Adjunct Lecturer
BRAC University, Dhaka, Bangladesh
Designed examinations, assessments and delivered lectures on Object-Oriented Programming, Algorithms, Artificial Intelligence, and Introduction to Robotics.
Faculty profile: cse.sds.bracu.ac.bd/faculty_profile/436/ayon_roy

05/2025–08/2025
Digitalyst Intern
AI Automation Department, Banglalink Digital Communications Ltd, Dhaka, Bangladesh
Developed automation tools, including a Python-based cell-monitoring and alert system, a 2G half-rate dimensioning prediction model using time-series data with a web interface, and scripts integrating Google Earth data into the Customer Complaint Automation back end.

03/2025–04/2025
Global Intern
Excelerate, Remote
Collaborated with international students on data cleaning using Python and SQL, exploratory data analysis (EDA), ETA reporting, the construction of master tables in PostgreSQL, and data visualisation using tools such as Looker Studio.

03/2025–06/2025
Academic Intern
Information and Communication Technology Division, NCC Bank PLC, Dhaka, Bangladesh
Worked with the machine-learning team to perform systematic data preprocessing and tune the parameters of a deep-learning-based fraud-detection model for mobile financial service fund transfers.

06/2023–09/2024
Undergraduate Student Tutor / Teaching Assistant
Department of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh
Supported students in programming, Discrete Mathematics, and problem-solving. Collaborated with faculty to assess assignments, quizzes, and manage grade records.

SKILLS

Research & teaching — Research methods, academic writing, student engagement, assessment, and feedback

AI & data — Natural language processing, large language models, algorithms, agents, image processing, and machine learning for time-series, text, and image data

Security — Cryptographic models and encryption and decryption techniques

Professional — Teamwork and leadership in multicultural industries and international settings

EXTRA-CURRICULAR ACTIVITIES

- Effective Leadership certification by HP LIFE
- Ex-Co-Advisor, BRACU E-Sports Club; Ex-Member, BRACU Computer Club

REFERENCES

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